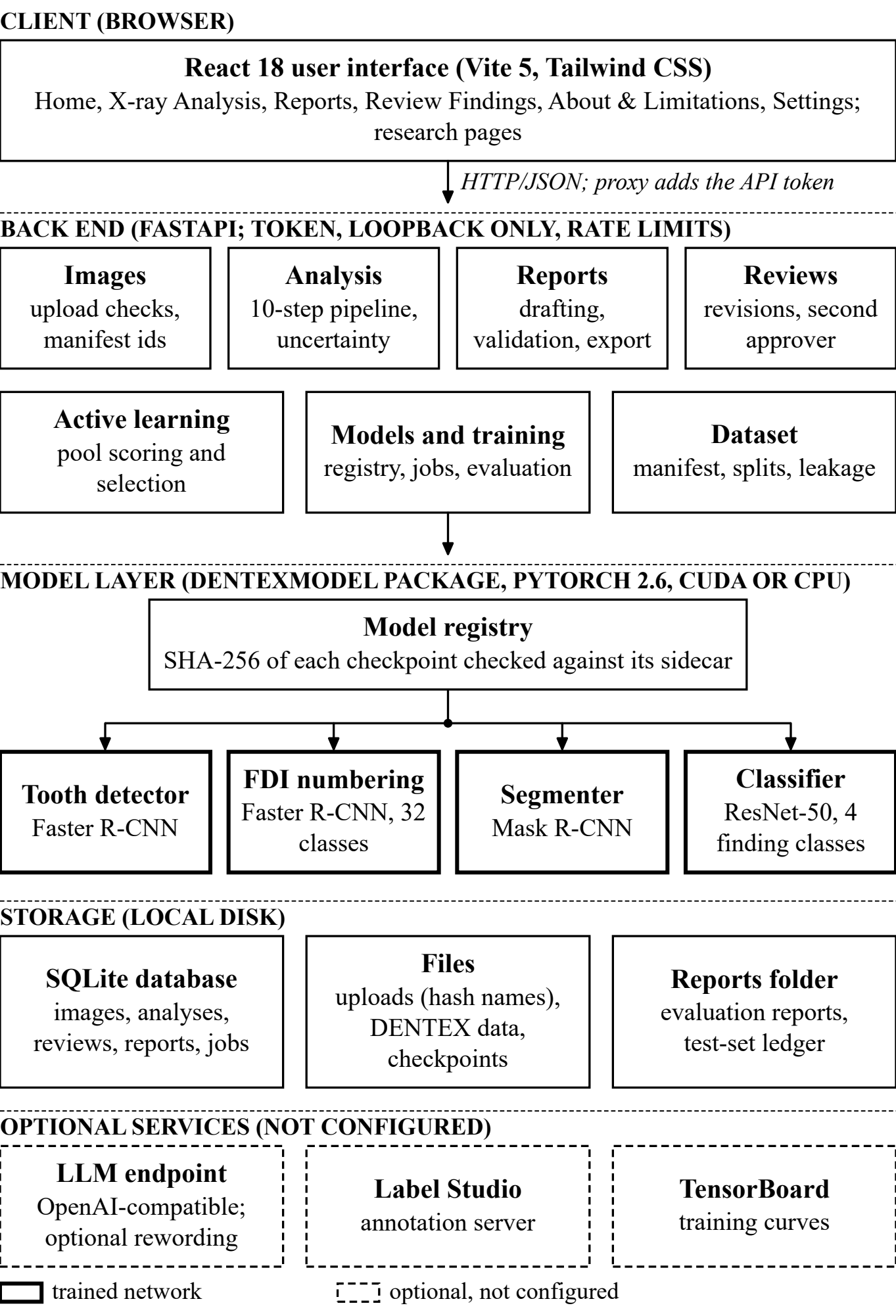

CLIENT (BROWSER)
React 18 user interface (Vite 5, Tailwind CSS)
Home, X-ray Analysis, Reports, Review Findings, About & Limitations, Settings; research pages
HTTP/JSON; proxy adds the API token
BACK END (FASTAPI; TOKEN, LOOPBACK ONLY, RATE LIMITS)
Images
upload checks, manifest ids
Analysis
10-step pipeline, uncertainty
Reports
drafting, validation, export
Reviews
revisions, second approver
Active learning
pool scoring and selection
Models and training
registry, jobs, evaluation
Dataset
manifest, splits, leakage
MODEL LAYER (DENTEXMODEL PACKAGE, PYTORCH 2.6, CUDA OR CPU)
Model registry
SHA-256 of each checkpoint checked against its sidecar
Tooth detector
Faster R-CNN
FDI numbering
Faster R-CNN, 32 classes
Segmenter
Mask R-CNN
Classifier
ResNet-50, 4 finding classes
STORAGE (LOCAL DISK)
SQLite database
images, analyses, reviews, reports, jobs
Files
uploads (hash names), DENTEX data, checkpoints
Reports folder
evaluation reports, test-set ledger
OPTIONAL SERVICES (NOT CONFIGURED)
LLM endpoint
OpenAI-compatible; optional rewording
Label Studio
annotation server
TensorBoard
training curves
trained network
optional, not configured